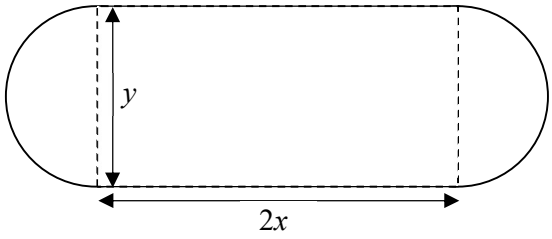
RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 1 (Delta)
Topic(s): Calculus I
Differentiation, Applications, Maclaurin and Graphing (Solutions)

1 [It is given that a sphere of radius r has surface area $4\pi r^2$ and volume $\frac{4}{3}\pi r^3$.]

A capsule is to be constructed using the curved surface of a cylinder and the curved surface of a hemisphere at each end. The length of the cylinder is $2x$ metres and the diameter of each hemisphere is y metres. The cross-sectional view of the capsule is shown in the diagram below.



It is given that the construction cost per square metre for the cylinder is \$3k and that for the hemispheres is \$5k, where k is a constant.

Given that the volume of the capsule is π m^3, show that the construction cost of the capsule, C satisfies $C = \frac{12\pi k}{y} + 3\pi k y^2$. [4]

Hence find, using differentiation, the exact values of x and y that give a minimum cost for its construction. [4]

Solution:

1

Surface area of cylindrical piece $= 2\pi\left(\frac{y}{2}\right)(2x) = 2\pi xy$

Surface area of 2 hemispherical pieces $= 2\times 2\pi\left(\frac{y}{2}\right)^2 = \pi y^2$

Let C be the construction cost and V be the volume of the capsule.

$C = 2\pi xy(3k) + \pi y^2(5k)$

$= \pi k y(6x + 5y)$

Since $V = \pi$,

$$\pi = \pi\left(\frac{y}{2}\right)^2(2x) + 2\times\frac{2}{3}\pi\left(\frac{y}{2}\right)^3$$

$$1 = \frac{xy^2}{2} + \frac{y^3}{6}$$

$$6 = 3xy^2 + y^3$$

$$x = \frac{2}{y^2} - \frac{y}{3}$$

$$C = \pi ky\left[6\left(\frac{2}{y^2} - \frac{y}{3}\right) + 5y\right]$$

$$= \pi ky\left(\frac{12}{y^2} - 2y + 5y\right)$$

$$= \frac{12\pi k}{y} + 3\pi ky^2 \text{ (shown)}$$

$$\frac{\mathrm{d}C}{\mathrm{d}y} = -\frac{12\pi k}{y^2} + 6\pi ky$$

Note that k is a constant

When $\frac{\mathrm{d}C}{\mathrm{d}y} = 0$,

$$\frac{12\pi k}{y^2} = 6\pi ky$$

$$y^3 = 2$$

$$y = \sqrt[3]{2}$$

When $y = \sqrt[3]{2}$, $x = \frac{2}{\left(\sqrt[3]{2}\right)^2} - \frac{\sqrt[3]{2}}{3} = 2^{\frac{1}{3}} - \frac{2^{\frac{1}{3}}}{3} = \frac{2}{3}\left(2^{\frac{1}{3}}\right) = \frac{1}{3}\left(2^{\frac{4}{3}}\right) = \frac{\sqrt[3]{16}}{3}$

$$\frac{\mathrm{d}^2C}{\mathrm{d}y^2} = \frac{24\pi k}{y^3} + 6\pi k$$

Need to show that the construction cost is minimised.

When $y^3 = 2$, $\frac{\mathrm{d}^2C}{\mathrm{d}y^2} = \frac{24\pi k}{2} + 6\pi k = 18\pi k > 0$ since $k > 0$

Hence, C is minimum when $y = \sqrt[3]{2}$ and $x = \frac{\sqrt[3]{16}}{3}$.

2 The curve C has parametric equations

$$x = t^2 + 2, \quad y = t^3, \text{ where } t \in \mathbb{R}.$$

(a) Sketch C. [1]

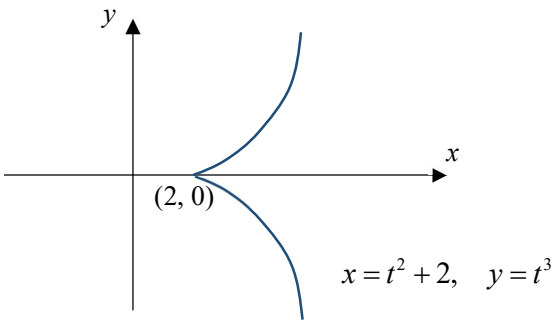
(b) Find the equation of the tangent to C at the point $(6, 8)$. [2]

(c) The tangent to C at the point $(6, 8)$ meets the curve C again at point P. Find the coordinates of point P. [3]

(d) The normal to C at the point $\left(m^2 + 2, m^3\right)$ meets the x- and y-axes at the points Q and R respectively. Given that the point F is the mid-point of QR, find the coordinates of F in terms of m. [4]

Solution:

2(a)



When $y = 0, \quad t = 0 \Rightarrow x = 2$

Note:
Set t in your GC to take both negative and positive values since $t \in \mathbb{R}$.

(b)

At $(6,8)$, $8 = t^3 \Rightarrow t = 2$

$\frac{dx}{dt} = 2t$ and $\frac{dy}{dt} = 3t^2$

$\frac{dy}{dx} = \frac{3t^2}{2t} = \frac{3}{2}t$

$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

When $t = 2$, $\frac{dy}{dx} = 3$

Equation of tangent is $y - 8 = 3(x - 6)$

$y = 3x - 10$.

Equation of a line with gradient m passing through $(k, \mathrm{f}(k))$:
$y - \mathrm{f}(k) = m\,(x - k)$

(c)

Since tangent to C at the point $(6,8)$ meets the curve C again at point P,

$t^3 = 3(t^2 + 2) - 10$

$t^3 - 3t^2 + 4 = 0$

It is not necessary to convert the parametric equations of curve C to cartesian equation.

Using GC,

$t = 2$ (reject as this gives point $(6,8)$) or $t = -1$

At $t = -1$, $x = (-1)^2 + 2 = 3$ and $y = (-1)^3 = -1$.

The coordinates of point P are $(3, -1)$.

For perpendicular lines with

(d)

At $\left(m^2+2, m^3\right)$, $t=m$, $\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{3m}{2}$

Gradient of the normal to the curve at $\left(m^2+2, m^3\right)$ is $-\dfrac{2}{3m}$

$y-m^3=-\dfrac{2}{3m}\left(x-m^2-2\right)$

i.e., $y=-\dfrac{2}{3m}x+\dfrac{2m}{3}+\dfrac{4}{3m}+m^3$

To find R: When $x=0$, $y=\dfrac{2m}{3}+\dfrac{4}{3m}+m^3$

To find Q: When $y=0$, $x=\dfrac{3m^4}{2}+m^2+2$

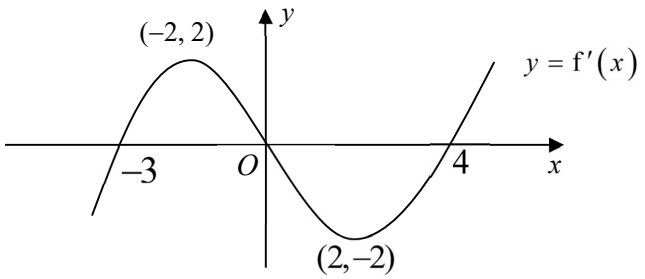
Coordinates of R and Q are $\left(0, \dfrac{2m}{3}+\dfrac{4}{3m}+m^3\right)$ and $\left(\dfrac{3m^4}{2}+m^2+2, 0\right)$.

$\therefore$ Coordinates of the mid-point F

$$=\left(\frac{\frac{3m^4}{2}+m^2+2}{2}, \frac{\frac{2m}{3}+\frac{4}{3m}+m^3}{2}\right)$$

$$=\left(\frac{3m^4}{4}+\frac{m^2}{2}+1, \frac{m}{3}+\frac{2}{3m}+\frac{m^3}{2}\right).$$

3 **(a)** The diagram shows the **derivative** graph of $y = \mathrm{f}(x)$.



Justifying your answers, find the range of values of x for which the graph $y = \mathrm{f}(x)$ is

(i) decreasing, [2]

(ii) increasing and concave downwards. [2]

(b) [It is given that the volume and surface area of a sphere of radius r is $\frac{4}{3}\pi r^3$ and $4\pi r^2$ respectively.]

Air is pumped into a spherical balloon at a constant rate of $12\pi\ \text{cm}^3$ per second.

(i) Find the rate of increase of the balloon's surface area when the volume is $\frac{256}{3}\pi\ \text{cm}^3$. [4]

(ii) Show that the rate of increase of the balloon's radius is inversely proportional to the surface area of the balloon. [1]

Solution:

<table>
<tr><td>3(a)(i)</td><td colspan="2">The graph of $y = \mathrm{f}(x)$ is decreasing when $\mathrm{f}'(x) < 0$.
Therefore, $x < -3$ or $0 < x < 4$.</td></tr>
<tr><td>(a)(ii)</td><td colspan="2">The graph of $y = \mathrm{f}(x)$ is increasing and concave downwards when $\mathrm{f}'(x) > 0$ and $\mathrm{f}'(x)$ is decreasing. Therefore, $-2 < x < 0$.</td></tr>
<tr><td>(b)(i)</td><td colspan="2">Let $V\ \text{cm}^3$ and $A\ \text{cm}^2$ denote the volume and surface area of the spherical balloon with radius r cm respectively.</td></tr>
<tr><td></td><td><u>Method 1</u>:
$V = \frac{4}{3}\pi r^3$
Differentiate w.r.t r
$\frac{\mathrm{d}V}{\mathrm{d}r} = 4\pi r^2$
Now, $\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{\mathrm{d}V}{\mathrm{d}r} \times \frac{\mathrm{d}r}{\mathrm{d}t}$
$12\pi = 4\pi r^2 \frac{\mathrm{d}r}{\mathrm{d}t}$</td><td><u>Method 2</u>:
$V = \frac{4}{3}\pi r^3$
Differentiate w.r.t t
$\frac{\mathrm{d}V}{\mathrm{d}t} = 4\pi r^2 \frac{\mathrm{d}r}{\mathrm{d}t}$
$12\pi = 4\pi r^2 \frac{\mathrm{d}r}{\mathrm{d}t}$
$\frac{\mathrm{d}r}{\mathrm{d}t} = \frac{3}{r^2}$</td></tr>
</table>

	$\therefore \frac{\mathrm{d}r}{\mathrm{d}t} = \frac{3}{r^2}$ $A = 4\pi r^2$ Differentiate w.r.t r $\frac{\mathrm{d}A}{\mathrm{d}r} = 8\pi r$ Now, $\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{\mathrm{d}A}{\mathrm{d}r} \times \frac{\mathrm{d}r}{\mathrm{d}t}$ $= 8\pi r\left(\frac{3}{r^2}\right)$ $= \frac{24\pi}{r}$ When $V = \frac{256}{3}\pi$, $\frac{4}{3}\pi r^3 = \frac{256}{3}\pi$ $r = 4$ When $r = 4$, $\frac{\mathrm{d}A}{\mathrm{d}t} = 6\pi \text{ cm}^2/\text{s}$	$A = 4\pi r^2$ Differentiate w.r.t t $\frac{\mathrm{d}A}{\mathrm{d}t} = 8\pi r \frac{\mathrm{d}r}{\mathrm{d}t}$ $= 8\pi r\left(\frac{3}{r^2}\right)$ $= \frac{24\pi}{r}$ When $V = \frac{256}{3}\pi$, $\frac{4}{3}\pi r^3 = \frac{256}{3}\pi$ $r = 4$ When $r = 4$, $\frac{\mathrm{d}A}{\mathrm{d}t} = 6\pi \text{ cm}^2/\text{s}$
(b)(ii)	From (bi), $\frac{\mathrm{d}r}{\mathrm{d}t} = \frac{3}{r^2}$ Since $A = 4\pi r^2$, $r^2 = \frac{A}{4\pi}$ $\frac{\mathrm{d}r}{\mathrm{d}t} = 3 \times \frac{4\pi}{A} = \frac{12\pi}{A}$ $\therefore$ The rate of increase of balloon's radius is inversely proportional to the area of the balloon (shown).	

4 The curve C has equation given by $y = \dfrac{1+3x-18x^2}{9x+3}$.

(a) Without the use of a calculator, find the range of values that y can take. [4]

(b) Sketch the graph of C indicating clearly its asymptotes and stationary points. [3]

(c) State a sequence of transformations that will transform the curve C to the curve with equation $y = \dfrac{1+3(2x-1)-18(2x-1)^2}{9(2x-1)+3}$. [2]

Solution:

4(a)

$$y = \frac{1+3x-18x^2}{9x+3}$$

$$(9x+3)y = 1+3x-18x^2$$

$$9xy+3y = 1+3x-18x^2$$

$$18x^2+(9y-3)x+3y-1=0$$

For the equation to have real roots,

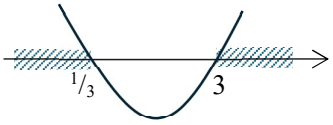
$$(9y-3)^2-4(18)(3y-1)\geq 0$$

$$9(3y-1)^2-4(18)(3y-1)\geq 0$$

$$9(3y-1)(3y-1-8)\geq 0$$

$$9(3y-1)(3y-9)\geq 0$$

$$27(3y-1)(y-3)\geq 0$$



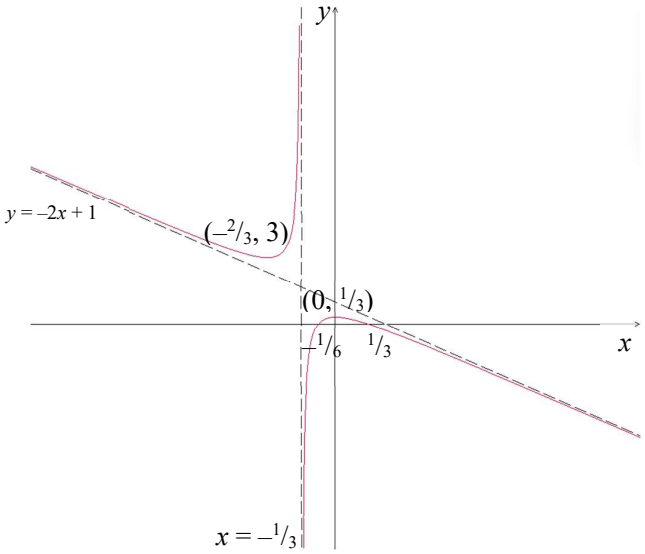
$$y \leq \frac{1}{3} \text{ or } y \geq 3$$

Hence the set required is $\{y \in \mathbb{R} : y \leq \frac{1}{3} \text{ or } y \geq 3\}$

(b)

$$\frac{1+3x-18x^2}{9x+3} = -2x+1-\frac{2}{9x+3}$$

$\therefore$ the asymptotes are $y=-2x+1$ and $x=-\frac{1}{3}$



(iii)

Let $y = \text{f}(x) = \frac{1+3x-18x^2}{9x+3}$

$$y = \text{f}(x) \quad \rightarrow \quad y = \text{f}(x-1) = \frac{1+3(x-1)-18(x-1)^2}{9(x-1)+3} \qquad \textit{Replace x by x-1}$$

$$\rightarrow \quad y = \text{f}(2x-1) = \frac{1+3(2x-1)-18(2x-1)^2}{9(2x-1)+3} \qquad \textit{Replace x by 2x}$$

The transformations are

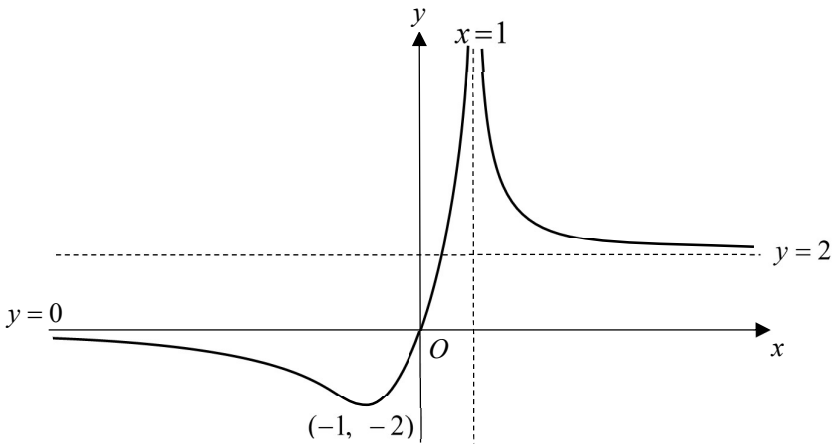
(1) Translation of the curve in the positive x direction by 1 unit.

(2) Scaling of scale factor $\frac{1}{2}$ along the direction of the x-axis.

OR: (1) Scaling of scale factor $\frac{1}{2}$ along the direction of the x-axis.

(2) Translation of the curve in the positive x direction by $\frac{1}{2}$ unit.

5 The graph of $y=\mathrm{f}(x)$ is given below. It has one vertical asymptote at $x=1$ and two horizontal asymptotes $y=0$ and $y=2$. The graph passes through the origin O and has a turning point at $(-1,\ -2)$.



(a) On separate diagrams, sketch the following graphs, indicating the coordinates of the points where the graphs cross the axes, the turning points, and the equations of any asymptotes if possible.

(i) $y=\dfrac{1}{\mathrm{f}(x)}$ [3]

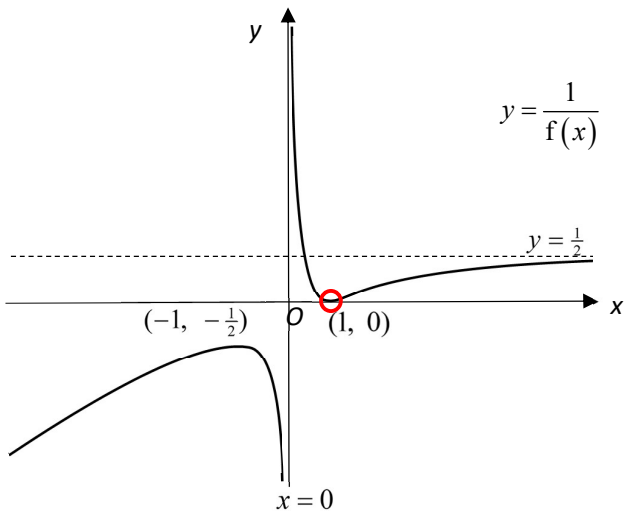
(ii) $y=\mathrm{f}'(x)$ [3]

(iii) $y=\mathrm{f}(|x-1|)$ [2]

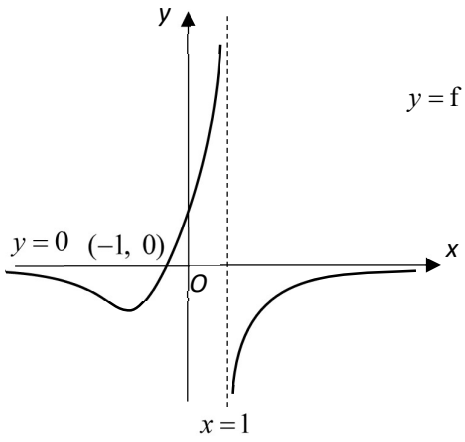
(b) State the range of values of a such that $\mathrm{f}(|x-1|)=a$ has only positive root(s). [1]

Solution:

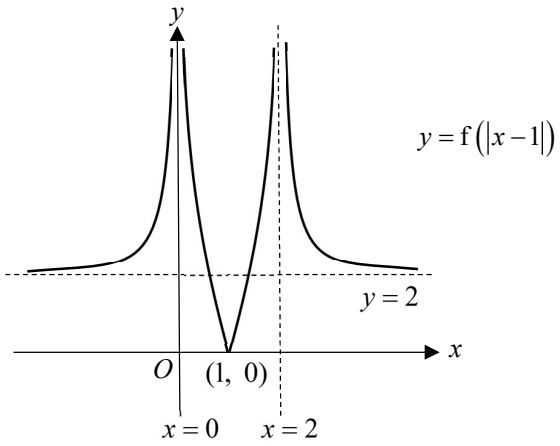
(a)
(i)



(ii)



(iii)



This graph is achieved in two steps in the order shown below:

1. Replace x by $|x|$
2. Replace x by $x - 1$

(b) $0 \le a \le 2$

Insert the graph of $y = a$ where it intersects the graph of $y = \mathrm{f}(|x-1|)$ on the right of the y-axis.

6 **(a)** State a sequence of transformations which maps the graph of $(x-1)^2+(y+1)^2=1$ to the graph of $\left(\frac{x-a}{a}\right)^2+(2y)^2=1$, where a is a positive constant. [3]

(b) The graph of $y=\mathrm{f}(x)$ has axial intercepts at $(a,0)$ and $(0,b)$, and an asymptote at $y=b$, where a and b are positive constants.

State, if it is possible to do so,

(i) the equations of any asymptote(s) and coordinates of axial intercept(s) for the graph of $y=\frac{1}{\mathrm{f}(x)}$, [3]

(ii) the coordinates of all axial intercept(s) for the graph of $y=\mathrm{f}(|x|)$. [2]

Solution:

6(a)

Method 1:

$(x-1)^2+(y+1)^2=1$

$\downarrow \quad x\to\frac{x}{a} \quad (1)$

$\left(\frac{x}{a}-1\right)^2+(y+1)^2=1$

$\downarrow \quad y\to y-1 \quad (2)$

$\left(\frac{x}{a}-1\right)^2+[(y-1)+1]^2=1$

$\Rightarrow\left(\frac{x-a}{a}\right)^2+(y)^2=1$

$\downarrow \quad y\to 2y=\frac{y}{1/2} \quad (3)$

$\left(\frac{x-a}{a}\right)^2+(2y)^2=1$

Note : $\left(\frac{x-a}{a}\right)^2=\left(\frac{x}{a}-1\right)^2$

(1) Scale parallel to x-axis by a factor of a.

(2) Translate 1 unit in the positive y-direction.

(3) Scale parallel to y-axis by a factor of $\frac{1}{2}$.

	Method 2: $(x-1)^2+(y+1)^2=1$ $\downarrow\ x\to\frac{x}{a}\quad(1)$ (1) Scale parallel to x-axis by a factor of a. $\left(\frac{x}{a}-1\right)^2+(y+1)^2=1$ $\downarrow\ y\to 2y\quad(2)$ (2) Scale parallel to y-axis by a factor of $\frac{1}{2}$. $\left(\frac{x}{a}-1\right)^2+(2y+1)^2=1$ $\downarrow\ y\to y-\frac{1}{2}\quad(3)$ (3) Translate $\frac{1}{2}$ unit in the positive y-direction. $\left(\frac{x-a}{a}\right)^2+\left[2\left(y-\frac{1}{2}\right)+1\right]^2=1$ $\Rightarrow\left(\frac{x-a}{a}\right)^2+(2y)^2=1$ Note: (1) can be applied independently of (2) and (3), eg (2) followed by (3) followed by (1) is another possible sequence.
(b) **(i)**	y-intercept: $\left(0,\frac{1}{b}\right)$ asymptotes: $y=\frac{1}{b},\ x=a$
(ii)	y-intercept: $(0,b)$ x-intercepts: $(a,0),(-a,0)$

7 Given that $y = \dfrac{\ln\sqrt{1-x}}{2+x}$, where $-1 \le x < 1$, show that

$$2y + 2(2+x)\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{1-x} = 0\,. \qquad [2]$$

(i) By further differentiation, find the Maclaurin's series for y up to and including the term in x^3. [5]

(ii) Verify that the same result can be obtained if the standard series expansions are used. [3]

(iii) By substituting $x = -1$ to your result, show that $\ln 2 \approx \dfrac{a}{b}$, where a and b are constants to be found. [2]

Solution:

7

$$y = \frac{\ln\sqrt{1-x}}{2+x}$$

$$y(2+x) = \frac{1}{2}\ln(1-x)$$

Differentiate w.r.t. x,

$$y + (2+x)\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{2(1-x)}$$

$$2y + 2(2+x)\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{1-x} = 0 \quad \text{(shown)}$$

Note :

$$\ln\sqrt{1-x} = \frac{1}{2}\ln(1-x)$$

(i) Differentiate w.r.t. x,

$$2\frac{\mathrm{d}y}{\mathrm{d}x} + 2\frac{\mathrm{d}y}{\mathrm{d}x} + 2(2+x)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \frac{1}{(1-x)^2} = 0$$

$$4\frac{\mathrm{d}y}{\mathrm{d}x} + 2(2+x)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \frac{1}{(1-x)^2} = 0$$

Differentiate w.r.t. x,

$$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 2(2+x)\frac{\mathrm{d}^3y}{\mathrm{d}x^3} + \frac{2}{(1-x)^3} = 0$$

When $x = 0,\ y = 0,\ \dfrac{\mathrm{d}y}{\mathrm{d}x} = -\dfrac{1}{4},\ \dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0,\ 4\dfrac{\mathrm{d}^3y}{\mathrm{d}x^3} + 2 = 0 \Rightarrow \dfrac{\mathrm{d}^3y}{\mathrm{d}x^3} = -\dfrac{1}{2}$

$y = \mathrm{f}(0) + x\mathrm{f}'(0) + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}'''(0)}{3!}x^3 + \cdots$, where $\mathrm{f}(x) = \dfrac{\ln\sqrt{1-x}}{2+x}$

$$= 0 + \left(-\frac{1}{4}\right)x + (0)\frac{x^2}{2!} + \left(-\frac{1}{2}\right)\frac{x^3}{3!} + \ldots$$

$$\approx -\frac{1}{4}x - \frac{1}{12}x^3$$

(ii)

$$y = \frac{\ln\sqrt{1-x}}{2+x}$$

$$= \left[\frac{1}{2}\ln(1-x)\right](2+x)^{-1}$$

$$= \left[\frac{1}{2}\ln(1-x)\right]\left[2\left(1+\frac{x}{2}\right)\right]^{-1}$$

$$= \frac{1}{4}\left[\ln(1-x)\right]\left(1+\frac{x}{2}\right)^{-1}$$

$$= \frac{1}{4}\left(-x-\frac{x^2}{2}-\frac{x^3}{3}+....\right)\left(1-\frac{x}{2}+\frac{x^2}{4}-\frac{x^3}{8}+...\right)$$

From Formulae List

$$= \frac{1}{4}\left(-x+\frac{x^2}{2}-\frac{x^3}{4}-\frac{x^2}{2}+\frac{x^3}{4}-\frac{x^3}{3}+....\right)$$

$$\approx -\frac{1}{4}x-\frac{1}{12}x^3 \text{ (verified)}$$

(iii)

Substituting $x=-1$ into result in (i),

$$\ln\sqrt{2} \approx -\frac{1}{4}(-1)-\frac{1}{12}(-1)^3$$

$$\frac{1}{2}\ln 2 \approx \frac{1}{3}$$

$$\therefore \ln 2 \approx \frac{2}{3} \text{ where } a=2,\ b=3$$

8 Given that θ is a sufficiently small angle, show that

$$\frac{1}{\sin 2\theta + \cos\theta} \approx 1 + a\theta + b\theta^2,$$

where a and b are constants to be determined. [4]

Solution:

8 Given that θ is a sufficiently small angle,

$$\frac{1}{\sin 2\theta + \cos\theta}$$

$$\approx \frac{1}{2\theta + 1 - \frac{\theta^2}{2}}$$

$$= \left[1 + \left(2\theta - \frac{\theta^2}{2}\right)\right]^{-1}$$

$$= 1 + (-1)\left(2\theta - \frac{\theta^2}{2}\right) + \frac{(-1)(-2)}{2!}\left(2\theta - \frac{\theta^2}{2}\right)^2 + \ldots$$

$$= 1 - 2\theta + \frac{\theta^2}{2} + (2\theta)^2 + \ldots$$

$$\approx 1 - 2\theta + \frac{9}{2}\theta^2 \quad \text{where } a = -2,\ b = \frac{9}{2}$$

Small angle approximations:

$$\sin x \approx x$$

$$\cos x \approx 1 - \frac{x^2}{2}$$